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MILITARY SPECIFICATION

GROUTING MACHINES (MUD JACKS), PNEUMATIC-TIRED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mobile, gasoline-engine-driven, pneumatic-tired, hydraulic pressure grouting machines for pumping a sand-cement slurry under concrete slabs and compacting around sewers and bridge abutments.

1.2 Classifications. The grouting machines shall be one of the following minimum sizes as specified (see 6.2):

Size 1 - 45 cubic feet per hour (cu ft/h).
Size 2 - 220 cu ft/h.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
O-E-760	- Ethyl Alcohol (Ethanol); Denatured Alcohol: and Proprietary Solvent.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
ZZ-H-451	- Hose, Fire, Woven Jacketed- Rubber- or Latex or Rubber Coated Fabric Lined, With Couplings.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-C-104	- Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-L-2105	- Lubricating Oil, Gear, Multi-purpose.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000 Pound Capacity.
MIL-E-10062	- Engines, Preparation for Shipment and Storage of.
MIL-C-21567	- Compound, Silicone, Soft Film.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing, Base.
MIL-P-46093	- Primer Coating, Synthetic (for Brake Drums).
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1474	- Noise limits for Army material.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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TIRE AND RIM ASSOCIATION, INC. (TRA)

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The grouting machine shall be capable of delivering, through the discharge nozzle, a mixture of slurry of various consistencies composed of topsoil screened through a 1/2-inch screen, sand, cement, and water. Size 1 shall have a capacity of not less than 45 cubic feet per hour at an operating discharge pressure of 100 pounds per square inch (lb/in^2). Size 2 shall have a minimum capacity of 220 cu ft/h at an operating discharge pressure of 250 lb/in^2. The grouting machine shall be a self-contained, wheel-mounted unit, complete and ready for operation. The grouting machine shall consist essentially of a slurry chamber, discharge pump, discharge hose with nozzle, and a gasoline engine power unit with a clutch. All necessary connections, gages, valves, and controls shall be furnished. In addition, the size 2 grouting machine shall include a water supply pump with suction hose and a mechanical agitator for mixing grout.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete grouting machine for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Maintainability. The grouting machine design shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional general-purpose tools associated with equipment of this nature. The replacement and adjustment of components shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the grouting machine.

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* 3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such a nature or so located as to be or become a hazard to the operating personnel shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard. The grouting machine shall comply with OSHA regulations in effect at time of manufacture.

3.6.1 Noise level. When specified (see 6.2), the noise level of the machine power unit, while operating at maximum rated capacity, shall be within limits for the preferred frequencies listed in MIL-STD-1474, category D. If the noise level is not within acceptable limits, the distance from the power unit at which it becomes acceptable shall be determined. That distance shall be shown on warning plates mounted on all four sides of the power unit, stating: "WARNING, EAR PROTECTION REQUIRED WITHIN _____FEET WHEN EQUIPMENT IS OPERATING". The warning shall be legible from the distance shown on the warning plate. The manufacturer of the grouting machine shall be responsible for filling in the number of feet for the required distance.

3.7 Standard commercial product. The grouting machine shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the grouting machine being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.8 Slurry chamber.

3.8.1 Size 1. The steel fabricated slurry chamber shall have a volume of not less than 4 cubic feet (cu ft), and shall be arranged to receive the premixed slurry directly and permit an unobstructed flow to the discharge pump.

3.8.2 Size 2. The steel fabricated slurry chamber shall have a volume of not less than 11 cu ft, equipped with a mixer of the continuous pugmill type, and shall have a paddle type mixing shaft with sealed roller or ball bearings. A two-way slurry discharge valve, equipped with dual connections and hose couplings or clamps for the connecting slurry hoses, shall be provided to facilitate changing or cleaning either one of the slurry hoses or discharge nozzle without stopping the entire operation. Water shall be supplied by means of a water pump on the machine. The mixer, water pump, grouting and discharge pump shall be driven by the power unit or individual power units.

3.9 Slurry discharge pump. The slurry discharge pump shall be of the low speed, vertical or horizontal, reciprocating type and shall be capable of maintaining the intended pressure and volume. Compression rings shall be of the rubber composition, replaceable type, of high abrasion-resistant quality.

3.9.1 Slurry valve. The slurry valve shall be of the free floating ball type with nonclogging action.

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3.9.2 Slurry discharge nozzle. The slurry discharge nozzle shall be a section of 2-1/2 inch outside diameter rubber hose of not less than four ply, designed for the intended use.

3.9.3. Slurry hose. Connection between the slurry pump and discharge nozzle shall be by means of a double-jacketed hose of the commercial type equal in quality to ZZ-H-451, 3 inches in diameter, 24 feet long, and include all necessary fittings.

3.10 Water pump. Size 2 grouting machine shall be equipped with a water supply pump and a check valve, capable of supplying water to the slurry chamber at a rate of not less than 20 gallons per minute. The water supply suction hose shall be of the high quality commercial type having an internal diameter of not less than 1-1/4 inches and be 24 feet long and equipped with a strainer and hose clamps or couplings.

3.11 Drives (size 2). The mixer for size 2 grouting machine shall be belt, chain, or gear driven by the power unit. The slurry discharge pump and water pump shall be driven by belt, chain, or gear from the mixer shaft or driven by individual power units.

* 3.12 Chassis. The grouting machine shall be mounted on a welded structural steel frame. The power unit and other parts requiring protection shall be enclosed. The size 1 grouting machine shall be of the wheelbarrow type, with two pneumatic-tired wheels, and have supporting leg(s) to provide stabilization during operation. It shall be movable on a level paved surface by one man. The size 2 grouting machine shall have two or four pneumatic-tired wheels, with either automotive or fifth-wheel type steering, capable of being towed at a speed of up to 25 miles per hour on paved roads, and shall be equipped with a retractable, telescopic, or swinging towbar with a lunette ring made from 1-5/8 inch round bar stock having an inside diameter of not less than 3 inches, and leveling jack with support shoe for stabilization.

3.12.1 Brakes and lights. The manufacturer's standard braking system shall be provided with automatic breakaway feature. Safety chains and stop, brake and directional lights shall be installed on the trailer.

3.13 Tires. Tires shall be of the tube or tubeless type, commercial quality, and shall have individual rated load-carrying capacities equal to the maximum individual rated loading imposed by the operation in accordance with T&RA recommendations.

3.14 Power unit. The grouting machine shall be powered by a gasoline engine having the horsepower, torque, and speed characteristics to satisfactorily meet all the grouting machine performance requirements specified herein.

3.14.1 Size 1. Size 1 grouting machine shall be powered by an air-cooled, single cylinder gasoline engine. The engine shall be furnished with a fuel tank having a capacity sufficient for not less than 4 hours continuous full load operation, a manual cranking system, and all the necessary accessories for efficient operation and control. A clutch shall also be furnished.

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3.14.2 Size 2. Size 2 grouting machine shall be powered by an air or liquid cooled gasoline engine. The engine shall be furnished complete with a fuel tank having sufficient capacity for not less than 4 hours continuous operation, an electric cranking system, battery, a battery charging generator or alternator having a rated capacity compatible to the system, a clutch, and all the necessary accessories for efficient operation and control. The battery shall be dry charged, a filling of electrolyte need not be furnished. The electric system shall be 12 volt.

3.15 Lubrication. Lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 lb/in2 or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.16 Radio noise suppressors. When specified (see 6.2), grouting machine shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the grouting machine shall be within the limits of SAE J551.

3.17 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.18 Instruction plates. The grouting machine shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.19 Identification marking plate. An identification marking plate of a durable material shall be furnished by the contractor. The contractor will be required to stamp the necessary data on the plate and affix the plate in a conspicuous place on each grouting machine. Nomenclature shall be "GROUT MACH, GED, (cap.) _____CU FT/H".

3.20 Tools. Any special tools required for operation of the grouting machine and its attachments shall be provided.

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* 3.21 Lifting and tiedown attachments. When specified (see 6.2), the grouting machine shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the grouting machine. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tying down attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the grouting machine on the carrier when shipped.

* 3.22 Workmanship.

3.22.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.22.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one grouting machine of the same size when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, and the packaging inspection of 4.5.

4.3 Examination. Each grouting machine shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall be tested in accordance with 4.4.1 through 4.4.3. Failure of the grouting machine to pass any test shall be the basis for rejection.

4.4.1 Operational and performance tests. The grouting machine and components shall be adjusted, lubricated, and otherwise serviced by the manufacturer preparatory to testing. Lubricants used shall be those recommended by the manufacturer for use in the ambient temperature and atmospheric conditions prevailing at the test site at the time of testing. The engine shall be started and warmed up for the period of time recommended by the manufacturer. The operator performing the tests shall be furnished by, and at the expense of, the manufacturer. The test procedure shall include, but shall not necessarily be limited to, the following:

a. The grouting machine shall be operated at normal operating speed for a period of 1 hour. A sufficient amount of slurry shall be available to

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assure continuous operation for the test period. The mixture shall be in accordance with 3.1. The slurry discharge nozzle shall discharge into a tank whereby the volume may be measured to assure compliance with 3.1.

b. Pressure gages shall be installed to measure the operating discharge pressure (see 3.1).

c. Each grouting machine shall be operated to the extent necessary to make all adjustments required for the proper operation of the completed unit. The machine will be operated for a period of time sufficient to detect any malfunction of the unit.

4.4.2 Test for electromagnetic interference characteristics. When electromagnetic interference characteristics are required, the grouting machine shall be tested to determine conformance to 3.16. In lieu of tests to determine conformance to SAE J551, the manufacturer may furnish a certification that the machine meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed. Nonconformance to 3.16 shall constitute failure of the test.

4.4.3 Lifting and tying down attachment tests. When required, the lifting and tying down attachments shall be tested to conform to 3.21.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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5.1.1.5 Unit protection. Each size 1 grouting machine shall be preserved method II. Openings in the equipment shall not be sealed, so that desiccated air may circulate. Each size 2 grouting machine shall be preserved method 1 with sealing of openings where required.

5.1.1.6 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.6.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.6.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.7 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062. Each size 1 grouting machine shall be preserved type II, submethod IIa. Each size 2 grouting machine shall be preserved type 1, method 1.

5.1.1.7.1 Engine housing. When specified (see 6.2), all vents, louvers, and similar openings in the engine housing shall be sealed with tape conforming to PPP-T-60, type IV. Any large openings shall be covered with waterproof barrier conforming to PPP-B-1055, class E-2, H-5, or M-1 and secured with tape specified herein.

5.1.1.8 Disk type clutches. Surfaces of clutch assemblies, such as clutch pressure plates, springs, exposed surfaces of disk hubs, and spline shafts, linkages, pins, arms, sleeves, throwout collars, and all metal facings within the housings shall be sprayed or fogged with a thin film (0.4-0.6 mil thick) of primer conforming to TT-P-664 or MIL-P-46093. Coating of composition facings shall be held to a minimum. Unpainted surfaces of control mechanisms not enclosed shall be coated with type P-1 preservative. The clutches shall be disengaged.

5.1.1.9 Brakes. Clean all surfaces as required to remove contaminants. Coat unpainted exterior surfaces of rods, levers, cables, clevises, linkage, and associated parts with type P-1 preservative. Remove brake drum covers and spray enclosed metallic components with primer TT-P-664 or MIL-P-46093. Spray coat brake drum friction surfaces with a very thin coat (0.5-0.6 mil thick) of primer. Coat friction surfaces of cams, adjusting screws, and anchor pins with type P-6 or P-7 preservative. Backoff all brakeshoe adjusters for maximum clearance between brake drum and brakeshoes. Do not apply coating of any type to contact surfaces of electric brakes. Adjust brakeshoes to provide maximum clearance to drum surfaces and lined surfaces. In a conspicuous location in the operator's compartment, attach a weatherproof warning tag marked, "Brakes preserved. Do not apply brakes when vehicle is being moved. Use tow bar or similar arrangement for moving vehicle. Adjust brakes before placing equipment into operation." Markings shall be applied with waterproof material.

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5.1.1.9.1 Hydraulic brakes. Remove and disassemble the brake cylinder. Clean the cylinders, pistons, springs, and other metallic surfaces normally in contact with hydraulic brake fluid with alcohol conforming to O-E-760. Flush hydraulic lines with alcohol. Follow this by cleaning process C-8. Dry by procedure D-1. Coat the cylinders, pistons, springs, rubber cups, rubber dust boots, and other internal metallic and rubber parts with silicone protector, MIL-C-21567. Fill the hydraulic system and the hydraulic fluid supply reservoirs with silicone brake fluid, MIL-B-46176. Bleed as necessary to assure the system is completely filled with fluid. After filling pressure bleeder reservoir, allow fluid to stand 3-5 minutes for air bubbles to escape, before replacing cap. In order to prevent air entrainment, if vehicle is manually bled, do not depress brake pedal over one-half of travel.

5.1.1.10 Pumps and valves. Interior surfaces of pumps and valves shall be coated with type P-10 preservative, grade 10, in a manner to insure thorough coating of all interior parts and surfaces. Where rubber packing rings are not removed from slurry pumps, care shall be exercised to prevent the preservative from coming in contact with the rubber components. Rubber packing rings removed shall be preserved in accordance with MIL-P-116, method IC-1. Openings into the pumps and valves shall be closed with barrier material conforming to MIL-B-121, type 1, grade A, class 2, secured with tape, or with caps or plugs.

5.1.1.11 Drive mechanisms.

5.1.1.11.1 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30, or 50) shall have the housing filled to the operating level with type P-10 preservative, grade as applicable. Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable. Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. Gear housings containing any of the above shall be identified with a weatherproof tag to indicate: "The housing is filled to the operating level with lubricant. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.11.2 Enclosed chains. Enclosed chains and chain housings shall be preserved and tagged as specified for enclosed gears.

5.1.1.11.3 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method 1C-1 or 1C-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation". Markings shall be applied to the tags with a waterproof material.

5.1.1.12 Hose and fittings. Ferrous metal hose fittings shall be coated with type P-1 preservative. The hose shall be coiled to the minimum safe diameter and the coils secured with ties evenly spaced.

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5.1.1.13 Wheels, axles, bearings, and tires. When wheels are removed, the tires shall be inflated to two-thirds of the specified operating pressure. The wheel nuts shall be replaced on the studs and the exposed threaded surfaces shall be coated with type P-1 preservative. When wheels and hubs are removed from axles or spindles, the interior bearing surfaces of the hubs and the bearings or bearing surfaces of the axles or spindles shall be coated with type P-6 preservative. The coated surfaces of the axles or spindles shall be wrapped and hub openings shall be covered with barrier material conforming to MIL-B-121, type 1, grade A, class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, type IV. Bearings removed from the wheels shall be coated with type P-6 preservative and preserved method 1A-8. When wheels are not removed, the tires shall be inflated to 10 pounds above the specified operating pressure.

5.1.1.14 Slurry chamber. Size 2 grouting machines shall have the opening in the slurry chamber covered with barrier material conforming to PPP-B-1055, class H-5, L-4, or M-1 or L-P-378, type I to act as a watershed against the penetration of rain or splashing water.

5.1.1.15 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.16 Technical publications. Technical publications for each piece of equipment shall be preserved method 1C-1 or 1C-3.

5.1.1.17 Consolidations. Disassembled and separately packaged components, tools, and publications for each grouting machine shall be consolidated in containers conforming to PPP-B-636, class weather-resistant for size 1 machines and PPP-B-601, overseas type or PPP-B-621, class 2, for size 2 machines. The contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186. Containers shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Level B. Grouting machines shall be packaged as specified for level A with the following modifications and exceptions:

5.1.2.1 Unit protection. Each size 1 grouting machine shall be preserved method 1 with the sealing of openings where required.

5.1.2.2 Engines. Each size 1 grouting machine shall be preserved type II, method 1.

5.1.3 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Size 1 machines. Each machine with consolidation containers shall be packed in a box conforming to PPP-B-601, overseas type; PPP-B-621, class 2; MIL-B-26195, type II, with plywood superstructure and rubbing strips; or, a

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crate conforming to MIL-C-104, type II (with top, sides, and ends assembled with nails). An inspection window shall be provided for packs 15 cubic feet or larger with an inspection door located in the container to allow access to the humidity indicator. Cushioning, anchoring, blocking, and bracing shall be in accordance with the container specification when provided or MIL-STD-1186 when not provided.

5.2.1.2 Size 2 machines. Each complete machine shall be prepared for mobile shipment. Consolidation containers shall be secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment. Support legs (landing gears) shall be used for parking and storage only. For shipment, the trailer front shall be secured and supported by the use of wood cribbing or blocking, as applicable.

5.2.2 Level B.

5.2.2.1 Size 1 machines. Each machine with consolidation containers shall be packed as specified for level A except that containers shall conform to PPP-B-601, domestic type; PPP-B-621, class 1; MIL-B-26195, type I with plywood superstructure and rubbing strips; or, MIL-C-3774, type 1 with shroud. Inspection windows and doors are not required.

5.2.2.2 Size 2 machines. Each machine shall be prepared as specified for level A.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The grouting machines covered by this specification are intended for raising concrete slabs hydraulically by injecting a slurry mixture through drilled holes in the slabs. Size 1 grouting machine is recommended for application, such as sidewalk, curb, and other small areas requiring a small volume of the slurry mixture. Size 2 grouting machine is recommended for general highway or street maintenance requiring large volumes of slurry mixture for the extensive slab areas.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. When first article is required for inspection and approval (see 3.2 and 6.4).
d. When noise level shall conform to MIL-STD-1474 (see 3.6.1).
e. When electromagnetic interference characteristics and test limits are required, (see 3.16).
f. Color of finish coat required (see 3.17).
g. When lifting and tying down attachments are required (see 3.21).
h. Level of preservation and level of packing required (see 5.1 and 5.2).

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* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience ony and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this documents based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Air Force - 99

Preparing Activity:

Navy - YD

Project No. 3895-0266

Review Activity:

Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

3895-0266

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-18763E	GROUTING MACHINES (MUD JACKS), PNEUMATIC-TIRED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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